

Design Exploration of Robotic In-Car Accessories for Semi-Autonomous Vehicles

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ABSTRACT

This video introduces Auze, a robotic rearview ornament designed to enhance the semi-autonomous driving experience at SAE Level 3 (see Figure 1). While Level 3 vehicles permit hands-free, eyes-off driving in specific scenarios, drivers must be ready to resume control when necessary [1]. Auze, as a robotic in-car accessory, utilizes kinesthetic and auditory cues in conjunction with the existing handover system to facilitate takeovers. In addition to outlining Auze's design and interaction, the video traces the design journey from ideation workshop to exploratory prototyping [2]. It also incorporates real data derived from the public demonstration of Auze, to underscore Auze's efficacy in augmenting driver preparedness for handover processes. Auze represents our initial step in our mission to design assistive systems that elevate the semi-autonomous driving experience by fostering trust between the driver and the vehicle. We hope that Auze will inspire researchers and designers to adopt diverse, experience-centered approaches in creating such technologies.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interactive systems and tools**;

KEYWORDS

Semi-autonomous vehicle, SAE Level 3, Robotic object, In-car assistant, In-car accessory, Handover, Takeover Request, Design Exploration, Ideation Workshop

ACM Reference Format:

Max Fischer, Jongik Jeon, Seunghwa Pyo, Shota Kiuchi, Kumi Oda, Kentaro Honma, Miles Pennington, and Hyunjung Kim. 2024. Design Exploration of Robotic In-Car Accessories for Semi-Autonomous Vehicles. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3613905.3649119>

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Figure 1: Auze, a robotic rearview mirror ornament, elevates the SAE Level 3 semi-autonomous driving experience through kinesthetic and auditory cues.

ACKNOWLEDGMENTS

This work was supported by the Japan Society for the Promotion of Science (JSPS) KAKENHI Grant Numbers 22H00246.

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- [2] Hyunjung Kim, Max Fischer, Jongik Jeon, Seunghwa Pyo, Kentaro Honma, Miles Pennington, Liu Hailong, Keisuke Shimono, and Toshihiro Hiraoka. 2024. Design Exploration of Robotic In-Car Accessories for Semi-Autonomous Vehicles. In Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction(HRI '24). ACM, New York, NY, USA, <https://doi.org/10.1145/3610978.3640596>.